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# Increasing MMR vaccination rates significantly reduces measles infections in Gaines County, Texas

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## SUMMARY

In early 2025, Texas experienced its largest measles outbreak in decades, with Gaines County reporting 413 of the state's 750 confirmed cases between January 1 and June 24. Despite an 82% MMR (measles, mumps, rubella) vaccination rate in the county, various factors contributed to the outbreak, including vaccine hesitancy after COVID-19, safety concerns, and philosophical or religious beliefs. My main hypothesis was that increasing MMR vaccination rates in a population will significantly reduce the total number and peak number of infected individuals. This study used an SIRV (Susceptible, Infected, Recovered, Vaccinated) model in Python, combined with data from the Texas Department of State Health Services, to simulate measles spread in Gaines County. Results predicted approximately 528 infections with an 82% vaccination rate. Increasing the rate to 85% would reduce infections to 261, and at 95%, the outbreak would last just 33 days with only 6 total cases. The analysis revealed a negative correlation between vaccination coverage and infection peak size. These findings support two key conclusions: (1) predictive modeling effectively simulates outbreak dynamics, (2) higher vaccination rates dramatically reduce disease spread.

## INTRODUCTION

Measles, also known as rubeola, is a highly infectious viral disease (1). It typically lasts about 10-14 days if no other disorders are present. This disease can lead to major complications and even death. It results in a generalized rash that can remain for up to 2 weeks. Other major symptoms include a high fever, fatigue, sore muscles, gastrointestinal issues, and a cough (2). The primary acquisition of this disease is through contact with infected individuals via airborne route. The respiratory droplets can remain airborne for up to 2 hours. Among unvaccinated people, just one infected person can transmit the infection to 90% of other people in close proximity (1). Anyone older than 20 years and those younger than 5, pregnant women, and individuals with a weakened immune system are at the highest risk of severe complications from measles which include pneumonia, blindness, encephalitis with brain injury, as well as death. Diagnosis of this disease is confirmed through blood tests (detect presence of immunoglobulin M antibodies), or measle virus isolation from nose or throat swabs. Although

there is no specific antiviral treatment option for infected individuals, experts recommend treatment with vitamin A to avoid major complications in children. There are 2 vaccines that can aid in the prevention of measles: MMR (measles, mumps, rubella – the most common vaccine) and MMRV (measles, mumps, rubella, varicella). Two doses of either vaccine are strongly recommended.

Measles became a disease of concern in the United States beginning in 1912 (3). During this decade, approximately 6000 deaths resulting from measles were reported annually. This trend would continue to grow until the 1960s. Each year, about 3-4 million infections were reported, 400-500 deaths were reported, and 1000 individuals suffered encephalitis. Blood samples were collected from infected individuals by John F. Enders and Dr. Thomas C. Pebbles. These two health experts were able to successfully isolate a measles strain in the blood sample of 13-year-old David Edmonston. This strain was transformed into a vaccine licensed in the US. In 1968, an even more efficient and less virulent measles vaccine was developed by Maurice Hilleman and his team (now included in the current MMR/MMRV vaccines). In 1981, the number of reported cases was 80% fewer than the previous year. This decrease would continue until 1989 when an outbreak among school-aged children began. This outbreak prompted the necessity of a second dose of the MMR vaccine. The second dose resulted in an even more significant decline in the number of annual reported cases and in 2000, Measles was officially declared eliminated in the US. However, sporadic outbreaks have continued to occur in under-vaccinated communities

Recently in January 2025, two new cases of measles were reported in Harris County, Texas. A few more cases then began being reported in Gaines County. By early February, 10 Measles cases had been reported. This would be the beginning of the state's largest Measles outbreak since 1992. As of June 24th, 2025, there have been a total of 750 cases reported in Texas, with 413 of them in Gaines County (4). Although the spread has drastically declined in May and June, concerns remain about the potential resurgence of this disease.

My main hypothesis was that increasing MMR vaccination rates in a population will significantly reduce the total number and peak number of infected individuals. The simulation was conducted using an SIRV model (Susceptible, Infected, Recovered, Vaccinated model) in Python. This model provided me with the dynamics of the disease and potential future impacts based on current trends. This model also allowed me to predict the number of total infected individuals, which was used to determine the impact of the vaccination rates on the spread of the disease.

## RESULTS

In this model, S represents the susceptible population, I the infected individuals, R the recovered, and V the vaccinated. The parameters β, γ, δ, and μ denote the transmission, recovery, vaccination, and loss-of-immunity rates, respectively.

According to the SIRV model line graph, if an 82% vaccination rate persists in Gaines County, the Measles outbreak will likely stabilize around the beginning of July 2025 (See figure 2). This is because the number of individuals recovering increases up until this time, while the number of susceptible individuals decreases. The number of currently infected individuals peaks around April (as was the case in reality) and slowly decreases until it reaches 0 towards the middle of the Summer. These trends can be seen in Gaines County right now, as the number of cases reported in June so far is 4, which is a significant decrease from April's 178. The difference between the number of susceptible individuals and recovered individuals is far greater throughout the 85% vaccination rate line graph than the 82% vaccination rate line graph (See Figures 2 and 3).

Changing the vaccination rate to 95% offers the most ideal scenario (See Figures 1 and 4). This vaccination rate results in an expected outbreak duration of 33 days and approximately 6 infected individuals. While increasing the vaccination rate from 82% to 85% decreased the total number of infected individuals while increasing the duration of the outbreak, a 95% vaccination significantly decreases both the expected number of infections and duration of the outbreak. More specifically, the total number of infected individuals decreased by over 97%, and the expected outbreak duration decreased by 352 days.

The graph in Figure 5 was used to calculate the gamma and beta values for the SIRV model. It consists of data from the first 4 months of the outbreak. There is an exponential relationship seen between the number of infected individuals and number of months during the first 4 months of the outbreak. However, as the outbreak continued, the number of cases rapidly declined and there was a negative exponential relationship seen between the two variables.

As seen from Figure 6, there is a negative exponential relationship between the rate at which susceptible individuals become vaccinated and the peak number of infected individuals. If the proportion of Gaines County's population that is vaccinated remains at 82%, then the peak number of individuals infected at a given time will be over 200. However, if this rate increases by 0.01 each month, then the peak number of infected individuals will be under 20. Although increasing the vaccination rate may also increase the duration of the outbreak, it will significantly decrease the number of infected persons.

## DISCUSSION

Despite the significant progress made in the elimination of measle by using highly effective vaccines, the percentage of children receiving the first dose of this vaccine has declined from 86% to 83% between 2019 and 2023. Additionally, the graphs created in this study show that the case count for this disease will increase exponentially during the first few months before it gradually declines (the Curve Fitting Model displays an exponential relationship between case count and time). The SIRV model showed that slightly increasing the number of individuals vaccinated in a population will decrease the number of infected individuals, though the outbreak may last longer (as seen in this study from the vaccination rate increase from 82% to 85%). This increase in duration is likely because increasing the vaccination rate will decrease the transmission rate among unvaccinated individuals, causing the virus to persist for a longer period as it gradually finds the remaining unvaccinated individuals or those with breakthrough infections. However, increasing the vaccination rate to 95% decreases both the expected number of infected individuals and total number of cases. A 95% vaccination rate is recommended to achieve herd immunity (8). This proportion of vaccinated individuals is crucial to protect those who refuse to get vaccinated or are unable to, as it decreases the presence of measles in a population, making it exponentially less likely that an individual contracts the disease.

The SIRV model used in this study yields similar results to the vaccination model published by epiEngage and CDC's measles outbreak model. EpiEngage's user friendly model employed an SEIRV model (Susceptible, Exposed, Infected, Recovered, Vaccinated). The transmission, latency, recovery, and vaccination are modeled stochastically over time. It simulates a range of possible cases and time course, given the input values of initial susceptible population size, population, and vaccination rate. Similar to my model, when slightly increasing the vaccination rate from 82% to 85%, the outbreak duration increases but the total number of infected individuals decreases. However, when simulating the results at a 95% vaccination rate, both outbreak duration and number of infected individuals decrease significantly (9). CDC's outbreak model also employed an SEIRV model. In addition to the parameters included in EpiEngage's model, CDC's outbreak model also considered the basic reproduction number (number of cases generated from a single case), latent period, infectious period, and vaccine efficacy. Thousands of simulations were conducted to ensure relatively accurate results. After the extensive analysis, it was concluded that by increasing the vaccination rate from about 44% to 93%, the infectious period could be reduced from 5 days to 3.8 days. It also showed that delaying vaccination by two weeks in a population with a 93% vaccination rate would increase the number of total infections by about 52%. Without any intervention, there would likely have

been 235 total reported measles infections. However, with the 93% vaccination rate, this number was reduced by about 75% to 58. Additionally, according to the model, the outbreak would have lasted up to 90 days without significant intervention. However, with a 93% vaccination rate and encouraging citizens to get vaccinated as early as possible, this duration was shortened to about 45 days. Both of these models support the idea that increasing the vaccination rate will significantly reduce the total number of infected individuals. They also show how the duration of the outbreak can be reduced by encouraging over 90% of a population to get vaccinated.

## METHODS

The first step of this project was acquiring the appropriate data. This was partially done through the Texas DSHS online case counter by county. I was able to find each county's total number of reported cases (4). I then narrowed my focus on Gaines County, since that is where most of the cases in the state were reported. The next step was to find the number of cases in Gaines County by month since January to predict the trajectory of the outbreak. I was able to find this number using DSHS Texas data (See Figure 8 for a graph of the data) (5).

With the acquisition of this data, the SIRV model could now be constructed using Python. This model is used in epidemiology to predict the spread of infectious diseases within a population. It has four main components: susceptible, infected, recovered, and vaccinated (6). Susceptible individuals are those who are unvaccinated and have not yet been in contact with the disease but could be soon. Infected describes individuals who currently have the disease. Individuals in the Recovered category are those who contracted the disease in the past and no longer have it. These individuals are no longer susceptible to the disease. Vaccinated individuals are those who have received a vaccine pertaining to the disease. The transmission rate (beta) is the rate at which susceptible individuals become infected after contacting an infected individual. The recovery rate (gamma) describes how quickly infected individuals recover from the disease. Lastly, the vaccination rate (nu) describes how quickly susceptible individuals become vaccinated.

The first equation describes how the number of susceptible individuals, S(t), changes over time. Susceptible individuals can either become infected through contact with infectious individuals or be removed from the susceptible group by vaccination. The term $-(\beta SI)/N$ represents the rate of new infections, where $\beta$ is the transmission rate per contact, S is the

number of susceptible individuals, I is the number of infectious individuals, and N is the total population. The term −vS accounts for susceptible individuals who are vaccinated at a rate v.

$$dS/dt = -((\beta SI)/N) - vS \quad \text{[Equation 1]}$$

The second equation models the change in the number of infectious individuals, I(t), over time. The first term, (βSI)/N, represents the inflow of newly infected individuals from the susceptible group, while the term −γI represents the outflow of individuals who recover from infection at rate γ. This balance determines whether the infection spreads or declines within the population.

$$dI/dt = ((\beta SI)/N) - \gamma I \quad \text{[Equation 2]}$$

The third equation captures the rate of change of the recovered population, R(t). Since recovery is assumed to confer lasting immunity, recovered individuals move permanently from the infectious class to the recovered class. The term γI therefore reflects the rate at which infected individuals recover per unit of time.

$$dR/dt = \gamma I \quad \text{[Equation 3]}$$

Finally, the fourth equation describes the dynamics of the vaccinated group, V(t). Individuals are added to this group when susceptible individuals receive a vaccine, represented by the term vS. The model assumes that vaccination provides complete and permanent immunity, meaning vaccinated individuals do not return to the susceptible class.

$$dV/dt = vS \quad \text{[Equation 4]}$$

Together, these four equations form the SIRV compartmental model, which tracks how infection, recovery, and vaccination interact over time in a constant-size population. This framework enables quantitative analysis of epidemic dynamics and evaluation of vaccination strategies under varying transmission and recovery conditions

These equations were essential in creating the SIRV Model function in Python. The initial population (N) of Gaines County was 23289. Since there were 6 reported cases of Measles in Gaines County in January, I set the initial number of infected individuals (I0) to 6.

Initial Recovered (R0) was set to 0 since none of these individuals recovered during January. The vaccination rate of Gaines County has remained around 82% for a few months, so I set the number of initial vaccinated individuals to 0.82 x N. The number of susceptible individuals is equal to N – I0 – R0 – V0. I needed to calculate the infection and recovery rate.

To do this, I created another program titled Curve Fitting Model. I0 and R0 remained the same, but this model did not account for the vaccination rate, so S0 increased. An SIR model was used to estimate the transmission and recovery rates. It took time, estimated transmission (0.2) and estimated recovery rates (0.1) as input. It then used the Euler method to calculate how many new infections and recoveries are expected to happen during small intervals for the duration of the given period. These calculated changes are used to update the values (S, I, R, new infected, new recovered) for each interval. It uses the rate of change from each small interval to build up an approximate trajectory of how S, I, and R evolve over the simulation period (see figure 5). The curve_fit function employed the non-linear least squares optimization algorithm Levenberg-Marquardt to find the values for beta and gamma, which were 1 and 0.1266, respectively. These values, along with the SIRV values, time, and vaccination rate, were used as input for the main SIRV model. Using the differential equations, this function returned the rate of change for S, I, R, and V. The odeint function was used as the core of the simulation. This function took the same input as the main SIRV model to numerically solve the system of the differential equations as defined in the SIRV model. A line graph was constructed (using matplotlib.pyplot) to visually show the changes in the rates of change. The number of days the outbreak would persist was also calculated. Starting from January, each day the number of active cases was greater than 1 (the I part of the SIRV model), the variable, days, would increase by 1. The day the number of currently infected individuals reached 0 is when the count would stop and the value of days would be printed out to the console. Additionally, a bar graph was constructed to show the changes in the number of monthly cases. In the Curve Fitting Model, a graph was constructed to show the relationship between time and number of infected individuals if the vaccination rate stayed at 0. Additionally, using the same SIRV model, a graph was constructed showcasing the relationship between vaccination rates and peak number of infected individuals. All initial values were the same as previously described. The only difference this time was that the vaccination rate changed from 0.00 to 0.01, incrementing by 0.0005 each time to obtain 20 different vaccination rates to be entered into the SIRV model function separately. An array, peak_infections, was used to store the peak number of infected individuals for each given vaccination rate. The beta and gamma values stayed constant throughout. As before, odeint was used to solve the differential equations over the given points

of time. A resulting 2D array variable was created, with each row representing a specific time and the columns corresponding to the S, I, R and V values. The array was transposed using Python's .T method, so I (infected individuals) was now a one-dimensional array, and its maximum value could be extracted. This process was repeated for each vaccination rate.

**REFERENCES**


1. Cleveland Clinic. "Measles (Rubeola)." Cleveland Clinic, 2025, https://my.clevelandclinic.org/health/diseases/8584-measles. Accessed 24 June 2025.
2. National Library of Medicine. "Overview: Measles." National Institutes of Health, Sept. 2023, https://www.ncbi.nlm.nih.gov/books/NBK564984/. Accessed 30 June 2025.
3. Centers for Disease Control and Prevention. "History of Measles." Centers for Disease Control and Prevention, 9 May 2024, https://www.cdc.gov/measles/about/history.html. Accessed 24 June 2025.
4. Texas Health and Human Services. "Measles Virus Measles Outbreak in Texas." Texas Health and Human Services, 2024, https://www.dshs.texas.gov/measles. Accessed 24 June 2025.
5. Texas Department of State Health Services. "Measles Outbreak in Gaines County, Texas." Texas Health and Human Services, 5 Feb. 2025, https://www.dshs.texas.gov/news-alerts/measles-outbreak-gaines-county-texas. Accessed 30 June 2025.
6. Schlickeiser, Reinhard, and Martin Kröger. "Key Epidemic Parameters of the SIRV Model Determined from Past COVID-19 Mutant Waves." Pathogens, vol. 3, no. 4, 2023, pp. 42–56. https://doi.org/10.3390/pathogens3040042.
7. Minta, A. A., et al. "Progress Toward Measles Elimination — Worldwide, 2000–2023." MMWR. Morbidity and Mortality Weekly Report, vol. 73, no. 45, 2024, pp. 1036–42. https://doi.org/10.15585/mmwr.mm7345a4.
8. Kuppalli, Krutika, and Trish M. Perl. "Measles in Texas: Waning Vaccination and a Stark Warning for Public Health." The Lancet Infectious Diseases, May 2025, https://doi.org/10.1016/S1473-3099(25)00162-8. Accessed 30 June 2025.
9. epiEngage. "epiENGAGE Measles Outbreak Simulator." Texas Advanced Computing Center, 2025, https://epiengage-measles.tacc.utexas.edu. Accessed 5 July 2025.
10. Masters, N. B., et al. "Real-Time Use of a Dynamic Model to Measure the Impact of Public Health Interventions on Measles Outbreak Size and Duration — Chicago, Illinois, 2024." MMWR. Morbidity and Mortality Weekly Report, vol. 73, no. 19, 2024, pp. 430–34. https://doi.org/10.15585/mmwr.mm7319a2.

11. Sinha, Aarush. “Gaines County Measles Outbreak SIRV Model Code.” GitHub, 2025, https://github.com/Aarush-ai288/Gaines-County-Measles-Outbreak. Accessed 30 June 2025.

**FIGURES**

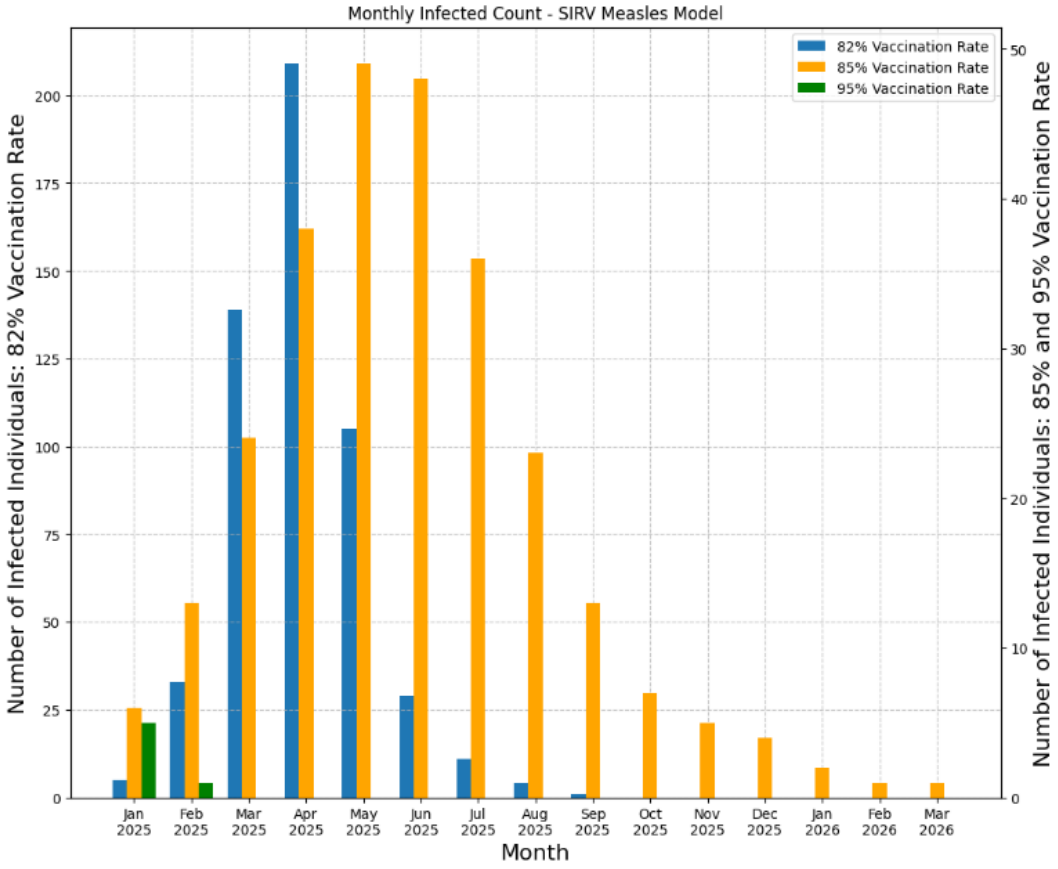



Figure 1. **Infected individuals per month for 82%, 85%, and 95% vaccination rates.** SIRV compartmental model used to simulate measles spread from Jan 2025–Mar 2026. No statistical tests; single simulation per scenario

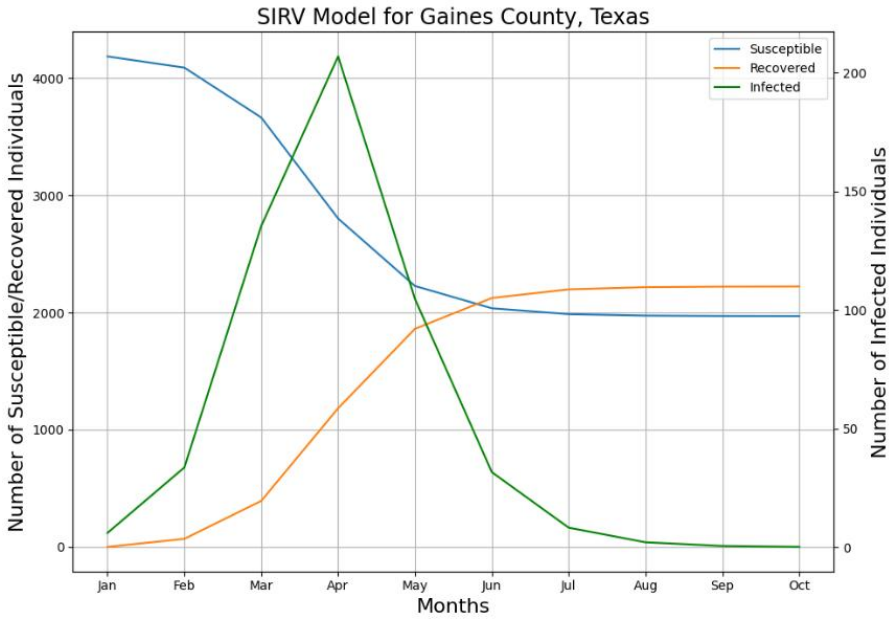



Figure 2. **Monthly trends of susceptible, infected, and recovered individuals from Jan to Oct at 82% vaccination rate.** SIRV compartmental model used to simulate measles spread. No statistical tests; single simulation run

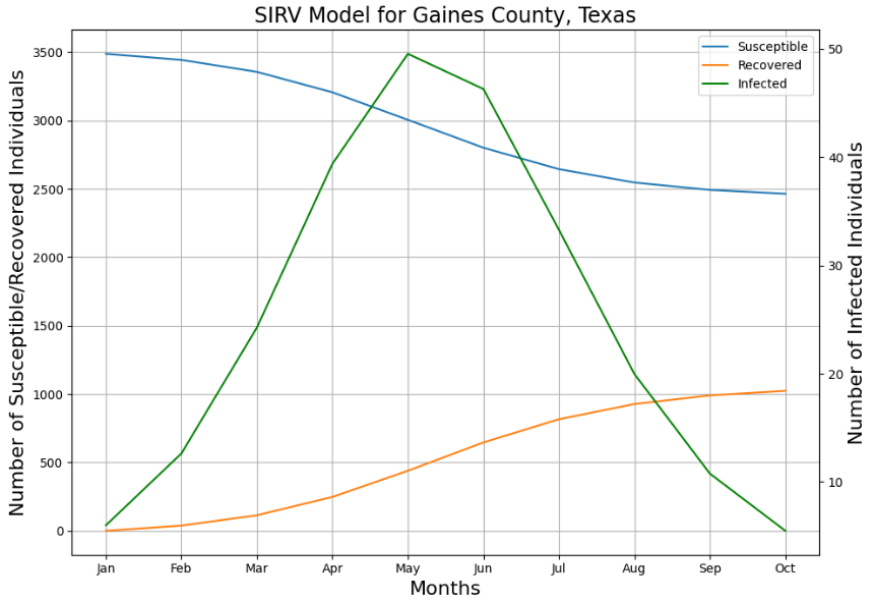



Figure 3. **Monthly trends of susceptible, infected, and recovered individuals from Jan to Oct at 85% vaccination rate.** SIRV compartmental model used to simulate measles spread. No statistical tests; single simulation run

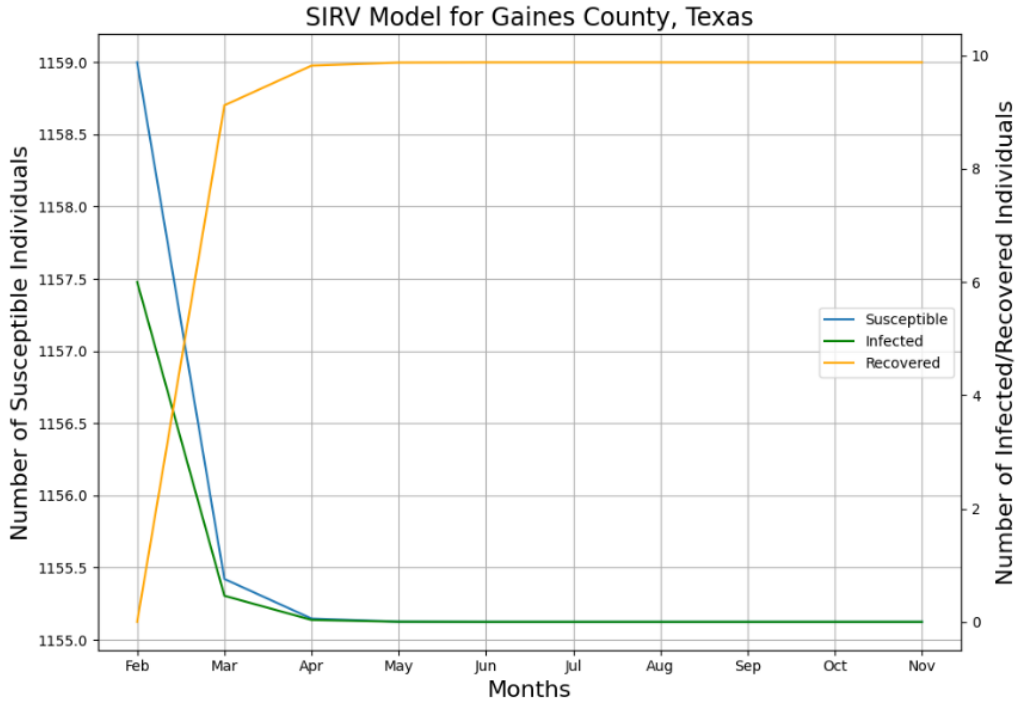



Figure 4. **Monthly trends of susceptible, infected, and recovered individuals from Feb to Nov at 95% vaccination rate.** SIRV compartmental model used to simulate measles spread. No statistical tests; single simulation run.

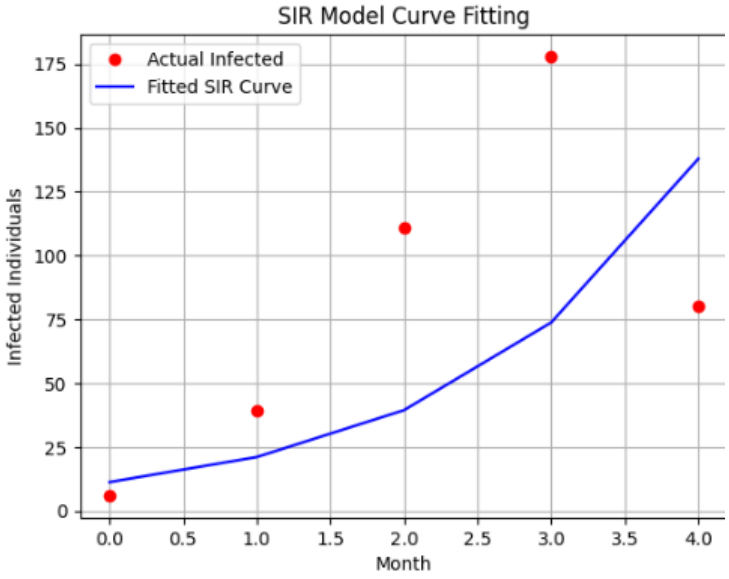



Figure 5. **SIRV Curve Fitting model.** Used to showcase the spread of measles in Gaines County, TX and calculate the beta and gamma values for the SIRV model. Regression line of best fit was used for this calculation. Single simulation run.

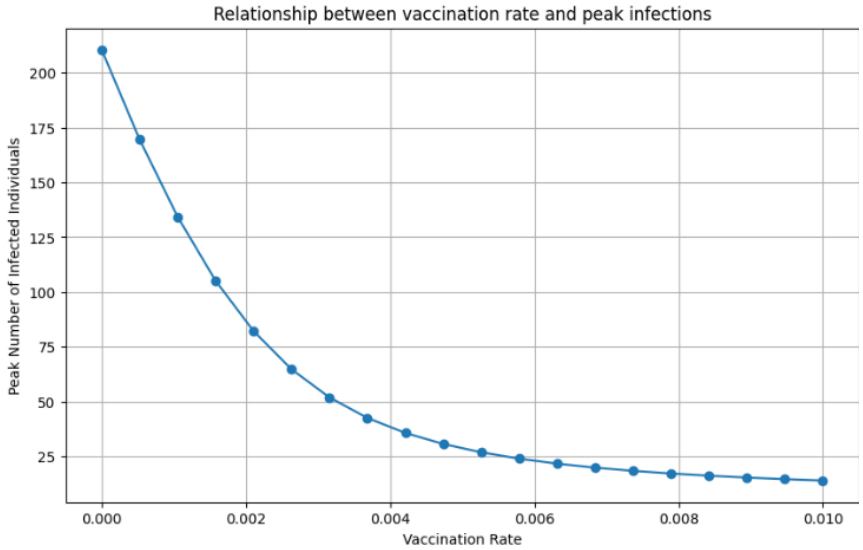



Figure 6. **Graph of Vaccination Rate and Peak Infections.** Data points generated from SIRV model. Used to show that the peak number of infected individuals will decline at higher vaccination rates. Line of best fit was used to show the relationship. Single simulation run.

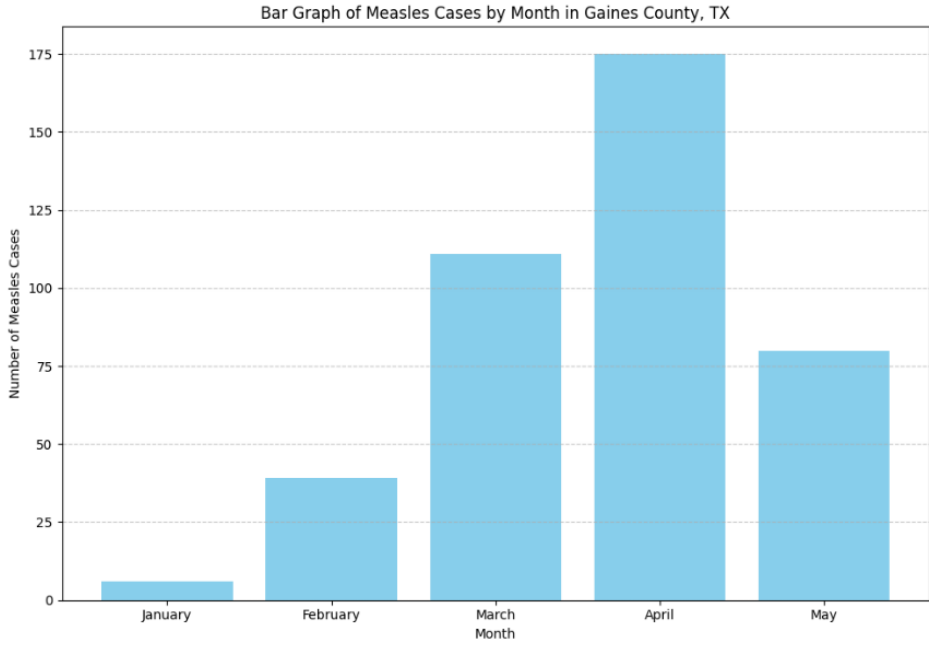



Figure 7. **Bar Graph of Measles Cases by Month in Gaines County, TX.** Measles cases by month in Gaines county from Jan 2025 to May 2025. Data found from CDC and Texas DSHS. Infections peak in April with approximately 175. Single simulation run.

## TABLES

| | 82% Vaccination Rate | 85% Vaccination Rate | 95% Vaccination Rate |
|---|---|---|---|
| **Total Predicted Number of Infected Individuals** | 528 | 261 | 33 |
| **Outbreak Duration (days)** | 266 | 385 | 6 |

Table 1: Total number of infected individuals and outbreak durations for 82%, 85%, and 95% vaccination rates for measles in Gaines County, TX. **Data found from CDC and DSHS.**